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VISUALIZATION WITH PYTHON

EMBARKING ON A JOURNEY
INTO DATA SCIENCE

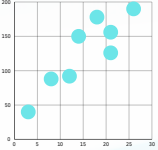
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DATA SCIENCE

You can have data without information but you
cannot have information without data.

-Daniel Keys Maran

DATA VISUALIZATION



In this section we'll cover key **data visualization** best practices for clear communication, with tips for choosing the right chart, formatting it effectively, and using it to tell a story

TOPICS WE'LL COVER:

Data Viz

3 Key Questions

Essential Visuals

Chart Formatting

Storytelling

Common Errors

GOALS FOR THIS SECTION:

- Understand the purpose behind visualizing data Learn the common chart types and their use cases
- Apply data visualization best practices to create clear and compelling charts
- Address common errors and how to avoid them

WHY VISUALIZE DATA?



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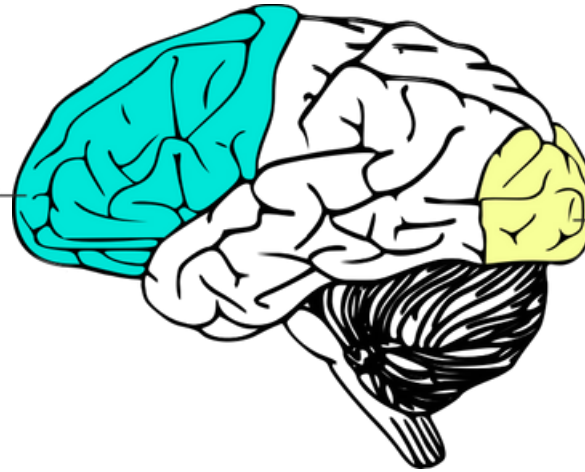
Common Errors

Data visualization is a way of presenting complex data in a form that is graphical and easy to understand.

- The human brain is built to interpret raw data as meaningless numbers and noise
- We need **clear patterns** and **visual cues** to help us quickly make sense of complex information

Prefrontal Cortex

- Located in the frontal lobe
- Responsible for cognitive functioning & problem solving
- Helps us make sense of non-visual information (like raw data)
- Slow & conscious



Visual Cortex

- Located in the occipital lobe
- Responsible for visual perception & understanding
- Helps us make sense of colors, patterns, shapes, sizes, etc.
- Instantaneous & subconscious

Data visualization puts both our **prefrontal** and **visual cortex** to work, combining the power of **cognition** (*slow and conscious*) and **perception** (*instantaneous*)

THE TEN SECOND RULE



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In **10 seconds**, what can you learn from the data below?

Coffee		Tea		Soda		Juice	
Units Sold	Profit	Units Sold	Profit	Units Sold	Profit	Units Sold	Profit
10	8.04	10	9.14	10	7.46	8	6.58
8	6.95	8	8.14	8	6.77	8	5.76
13	7.58	13	8.74	13	12.74	8	7.71
9	8.81	9	8.77	9	7.11	8	8.84
11	8.33	11	9.26	11	7.81	8	8.47
14	9.96	14	8.1	14	8.84	8	7.04
6	7.24	6	6.13	6	6.08	8	5.25
4	4.26	4	3.1	4	5.39	19	12.5
12	10.84	12	9.13	12	8.15	8	5.56
7	4.81	7	7.26	7	6.42	8	7.91
5	5.68	5	4.74	5	5.73	8	6.89

THE TEN SECOND RULE



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What if you were given the **averages**?

Coffee		Tea		Soda		Juice	
Units Sold	Profit	Units Sold	Profit	Units Sold	Profit	Units Sold	Profit
10	8.04	10	9.14	10	7.46	8	6.58
8	6.95	8	8.14	8	6.77	8	5.76
13	7.58	13	8.74	13	12.74	8	7.71
9	8.81	9	8.77	9	7.11	8	8.84
11	8.33	11	9.26	11	7.81	8	8.47
14	9.96	14	8.1	14	8.84	8	7.04
6	7.24	6	6.13	6	6.08	8	5.25
4	4.26	4	3.1	4	5.39	19	12.5
12	10.84	12	9.13	12	8.15	8	5.56
7	4.81	7	7.26	7	6.42	8	7.91
5	5.68	5	4.74	5	5.73	8	6.89
9	7.50	9	7.50	9	7.50	9	7.50

THE TEN SECOND RULE



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Data Viz

3 Key Questions

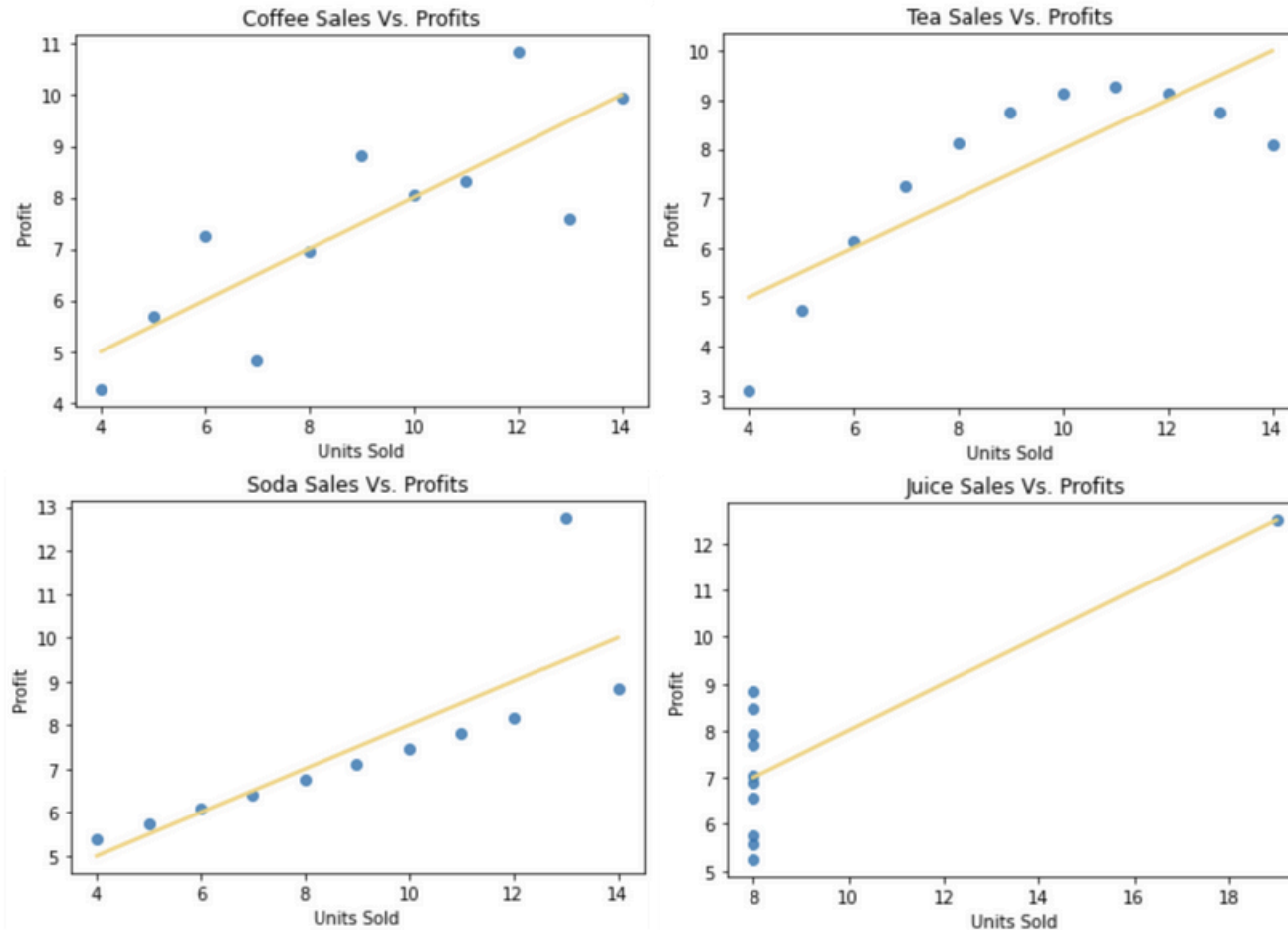
Essential Visuals

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What if you **visualize** it?



This is a slight twist on **Anscombe's Quartet**

Despite sharing nearly identical descriptive stats, each series tells a very different visual story



THREE QUESTIONS

THREE KEY QUESTIONS



1

What **TYPE OF DATA** are you working with?

Categorical? Numerical? Geospatial? Time-series? Hierarchical? Financial?

2

What do you want to **COMMUNICATE**?

Comparison? Composition? Relationship? Distribution?

3

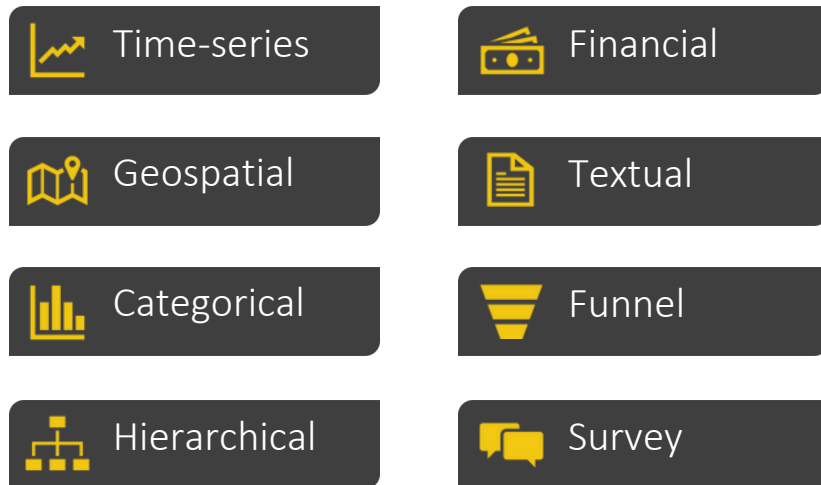
Who is the **END USER** and what do they need?

Analyst? Manager? Executive? General public?



THREE KEY QUESTIONS

1 What **TYPE OF DATA** are you working with?



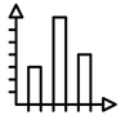
The type of data you're working with often determines **which type of visual will best represent it**; for example, using maps to represent geospatial data, line charts for time-series data, or tree maps for hierarchical data



THREE KEY QUESTIONS

2 What do you want to **COMMUNICATE**?

COMPARISON



Used to **compare values over time or across categories**

Common visuals:

- Column/Bar Chart
- Clustered Column/Bar
- Data Table/Heat Map
- Radar Chart
- Line Chart (*time series*)
- Area Chart (*time series*)

COMPOSITION



Used to **break down the component parts of a whole**

Common visuals:

- Stacked Bar/Column Chart
- Pie/Donut Chart
- Stacked Area (*time series*)
- Waterfall Chart (*gains/losses*)
- Funnel Chart (*stages*)
- Tree Map/sunburst (*hierarchies*)

DISTRIBUTION

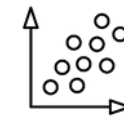


Used to **show the frequency of values within a series**

Common visuals:

- Histogram
- Density Plot
- Box & Whisker
- Scatter Plot
- Data Table/Heat Map
- Map/Choropleth (*geospatial*)

RELATIONSHIP



Used to **show correlation between multiple variables**

Common visuals:

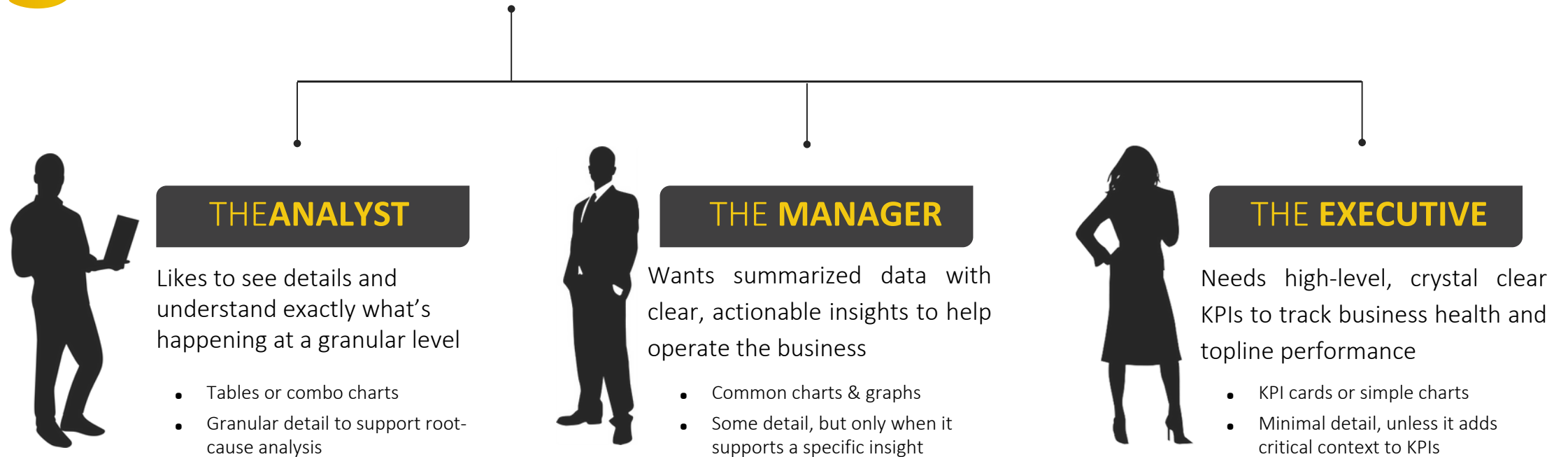
- Scatter Plot
- Bubble Chart
- Data Table/Heat Map
- Correlation Matrix

Keep it simple! While there are *hundreds* of charts to choose from, basic options like bars and columns, line charts, histograms and scatterplots often tell the simplest and clearest story

THREE KEY QUESTIONS



3 Who is the **END USER** and what do they need?



How you visualize and present your data is a function of **who will be consuming it**; a fellow analyst may want to see granular details, while managers and executives often prefer topline KPIs and clear, data-driven insights



ESSENTIAL VISUALS

KPI CARD

67%

Sometimes
simple text
works best

PIE CHART



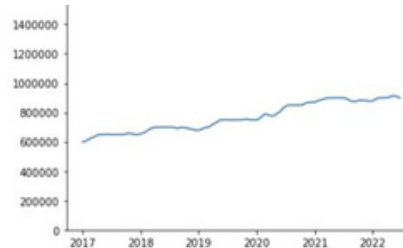
Sort the slices, keep
them under ~5, and
focus on one

TABLE

V2	4722.41	5690.51	5311.06
V51	1845.66	2034.86	4034.18
V52	2649.07	2968.23	5142.4
	H	I	J

Add a color scale to
highlight patterns in
the data

LINE CHART



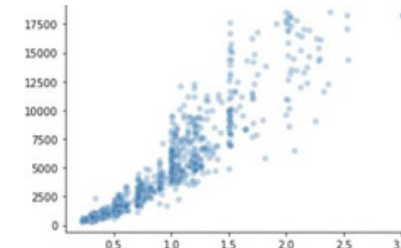
The dates must be
continuous

BAR CHART



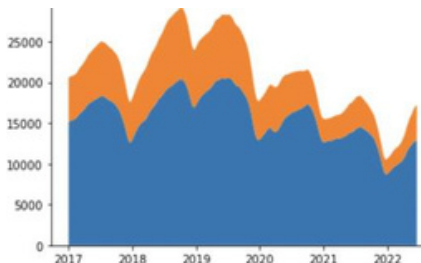
Baseline must start at zero

SCATTER PLOT



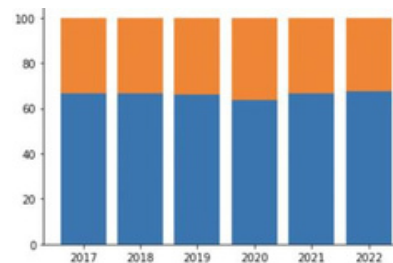
Remember that
correlation does not
imply causation

AREA CHART

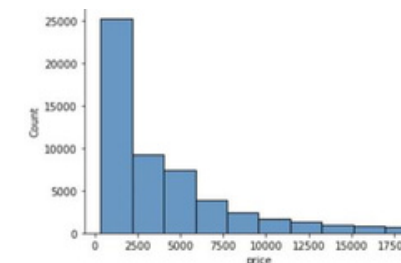


Comparison &
composition

100% STACKED



HISTOGRAM



Avoid using too
many bins!

CHART FORMATTING



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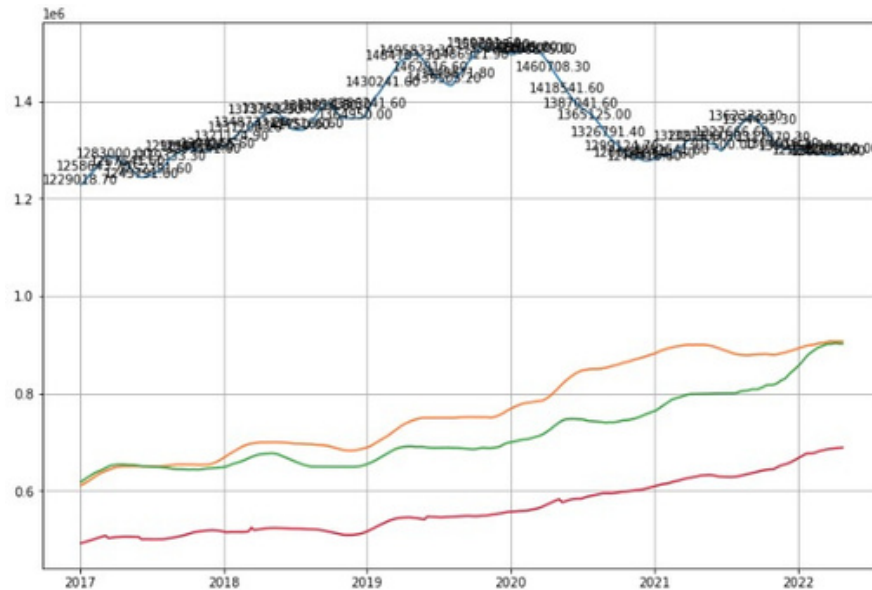
Formatting

Storytelling

Common Errors

Chart formatting should be used to eliminate noise & facilitate understanding

BEFORE: Cluttered chart



This is the right chart type... so why is it so hard to understand the visual?

- ✗ The chart border and gridlines are more distracting than useful
- ✗ The vertical axis labels are hard to read and lack context –it's using scientific notation and doesn't start at 0
- ✗ Data labels can help add context, but they just add noise here
- ✗ It's not clear what each line represents



Be intentional about the formatting you apply –don't just use the default settings!

CHART FORMATTING



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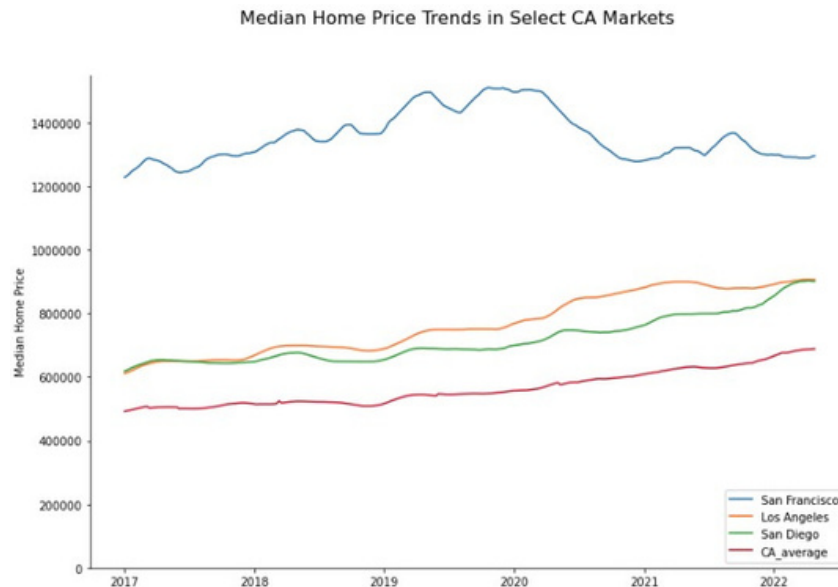
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Chart formatting should be used to eliminate noise & facilitate understanding

AFTER: Clear chart



PRO TIPS:

- ✓ Remove the chart border & gridlines
- ✓ Format the axis labels clearly
- ✓ Add context with the chart title
- ✓ Create a visual order
- ✓ Make sure the story is clear

*“Perfection is achieved not when there is nothing more to add, **but when there is nothing left to take away**”*

Antoine de Saint-Exupery

STORYTELLING



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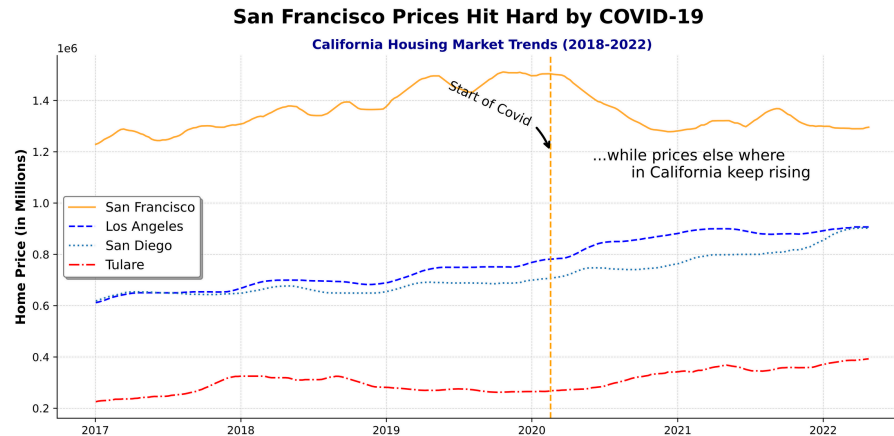
Formatting

Storytelling

Common Errors

Descriptive titles and data labels can be used **to tell a clear story** within your visuals

AFTER: Compelling chart



PRO TIPS:

- ✓ Leverage the title to guide the audience toward specific insights
- ✓ Insert text & shapes directly inside the chart
- ✓ Use data labels and annotations to draw attention to the main data points
- ✓ Use color strategically

COMMON ERRORS



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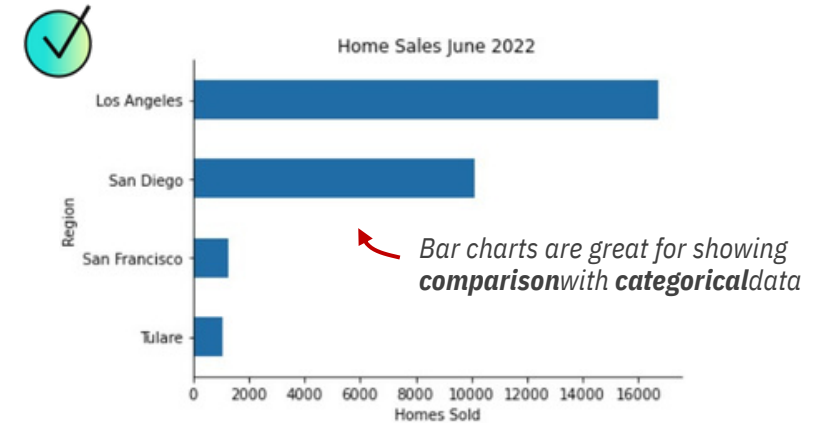
Common Errors

1

Choosing the **wrong visual** to represent the type of data



Using a line chart, which is meant for **time series data**, with categorical data gives the false sense of a trend



Bar charts are great for showing **comparison with categorical data**



While a tree map can work, comparisons and compositions are harder to make than with a bar or pie chart

It's best to use them with **hierarchical data**



PRO TIP: Don't prioritize variety over effectiveness; use the right chart for the job!

COMMON ERRORS



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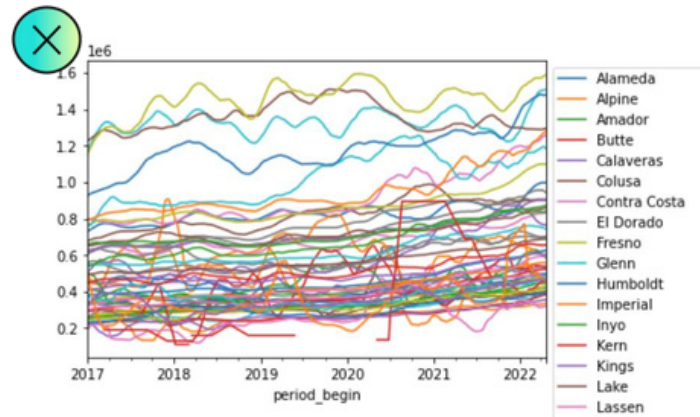
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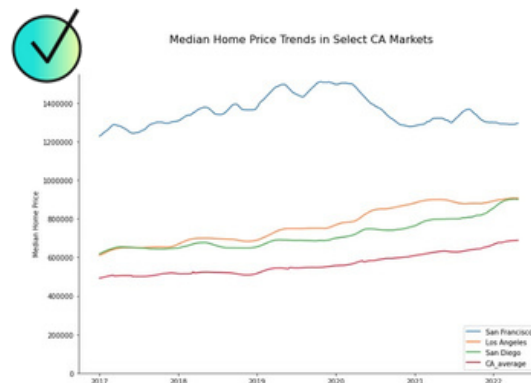
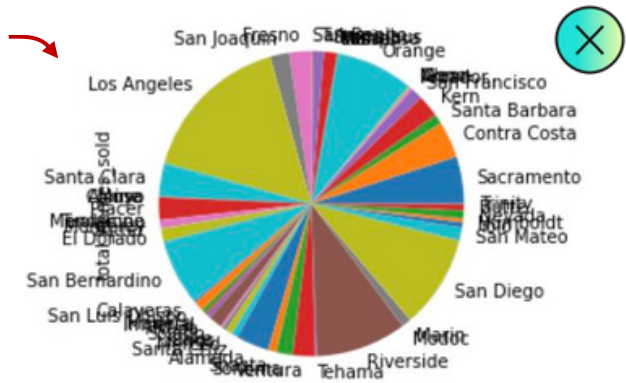
Storytelling

Common Errors

2 Including **too many series** in a single visual

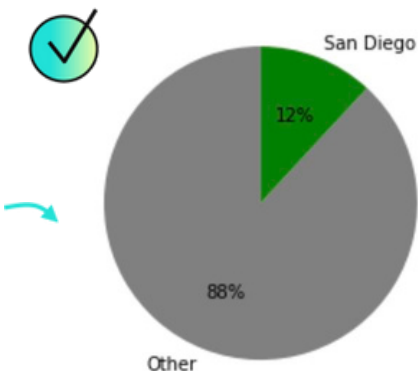


It's hard to focus or extract any valuable information



Try **highlighting** the series you want, or **aggregating** other categories

You can also **group** the other categories into a single series



COMMON ERRORS



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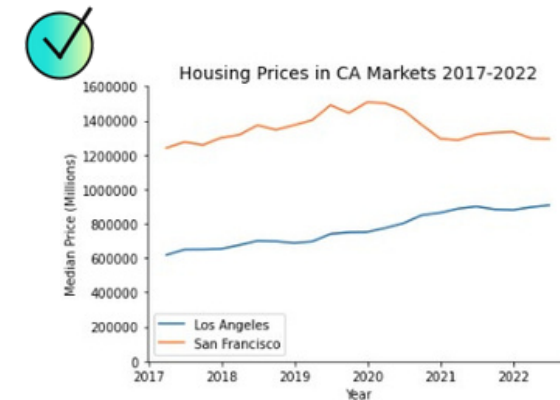
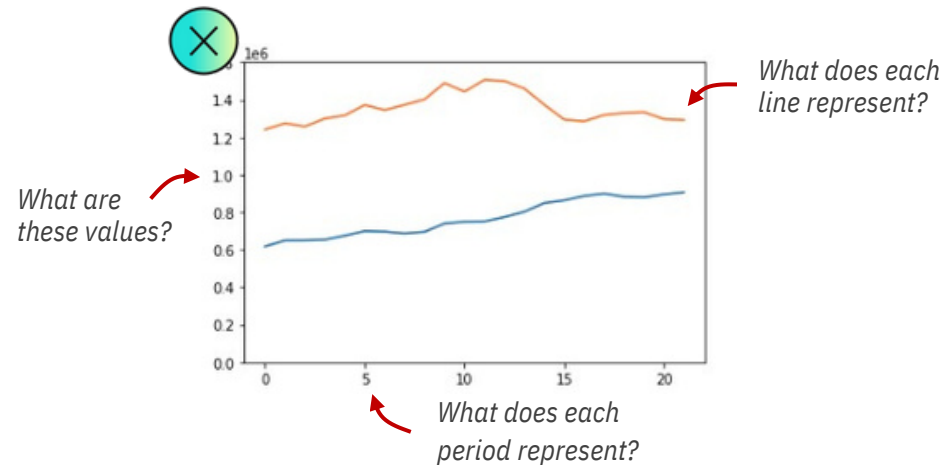
Formatting

Storytelling

Common Errors

3

Providing **little to no context** with text and labels



When removing elements from a chart to reduce clutter and noise, remember to **keep all the elements that add understanding**

COMMON ERRORS



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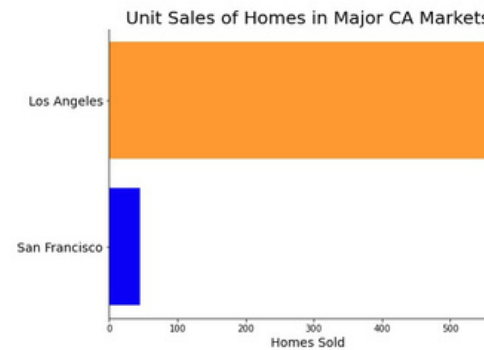
Essential Visuals

Formatting

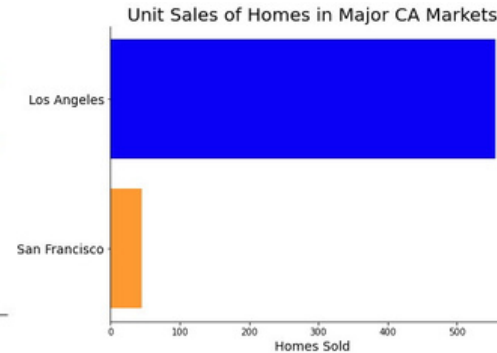
Storytelling

Common Errors

4 Using **inconsistent colors** between related visuals



Using **different colors for the same series** makes it difficult to associate them visually



Using **the same colors consistently** makes them easier to understand, and in some cases allows you to remove the legend



Consistency gains more importance **as the number of visuals increases**, making it critical for dashboards

KEY TAKEAWAYS



Always answer the **3 key questions** to choose the right visual

- *What type of data are you working with? What do you want to communicate? Who is the end user?*



Do **NOT** prioritize variety over effectiveness

- *Choose chart types based on how clearly they communicate the data underneath –you can customize later!*



Eliminate noise and distractions to **facilitate understanding**

- *“Perfection is achieved not when there is nothing more to add, but when there is nothing left to take away”*



Tell a story with the data to guide the user to the insights

- *Use titles, strategic labels, and callouts to create a clear narrative*